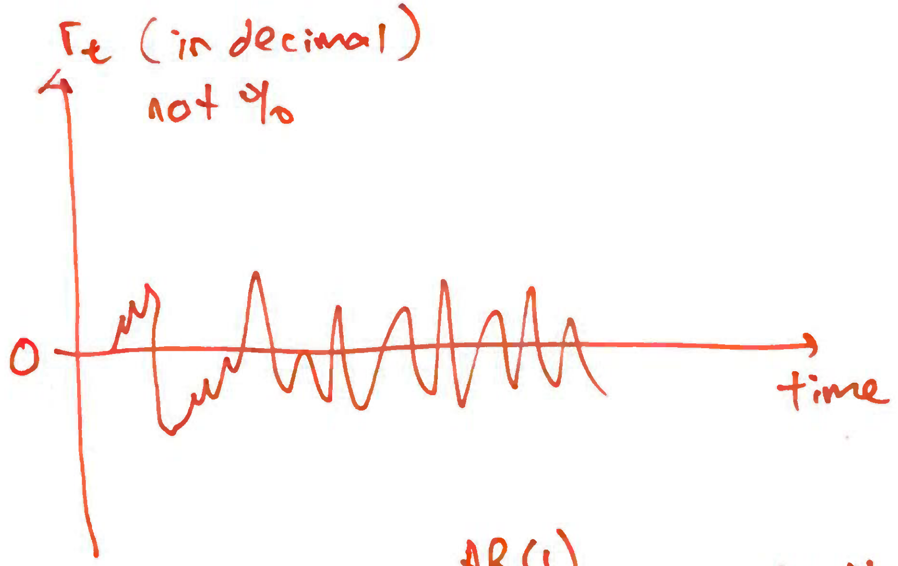
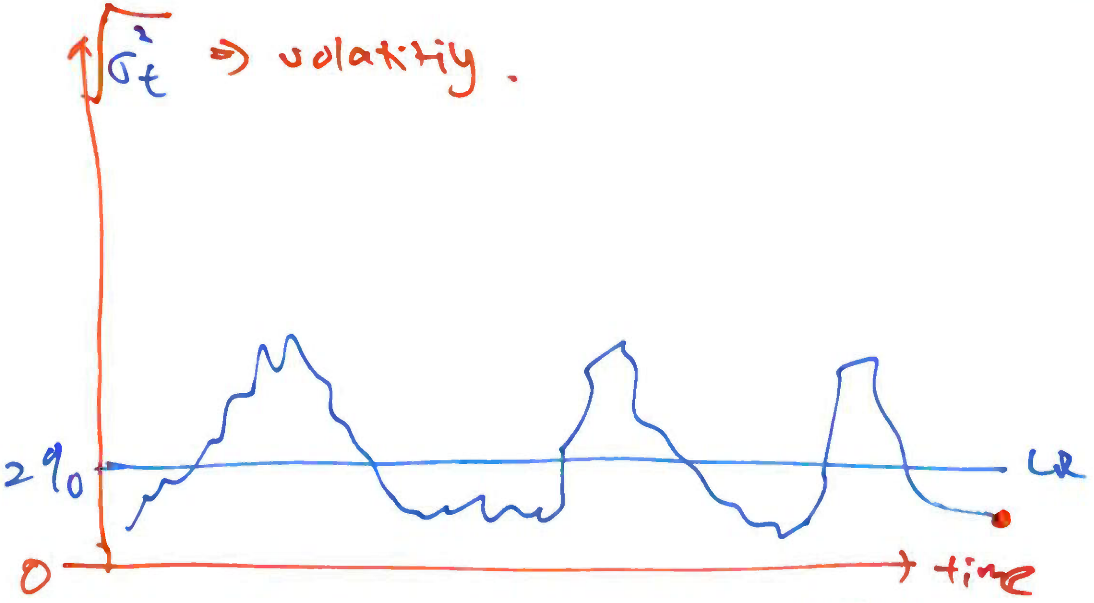




AR(1) → almost zero

WF, Slide 42

① $\mu_t = E(r_t \mid F_{t-1}) = c + \phi_1 r_{t-1}$

↓ best guest $\quad = SR$

$= R.V.$

① $\sigma_t^2 = V(\varepsilon_t \mid F_{t-1}) = SR \Rightarrow R.V.$

$= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$

wow

② $E(r_t) = \dfrac{c}{1-\phi_1} =$ a number $= 0$

$= LR \Rightarrow$ sample mean of r_t is about zero

② $\sqrt{V(\varepsilon_t)} = \sqrt{\dfrac{\alpha_0}{1-(\alpha_1+\beta_1)}} =$ a number

$= LR \qquad = 2\%$

LIE $\Rightarrow E(R.V.)$ ① $c = 0$ ② $\alpha_1 + \beta_1 < 1$ observed

MA(1)

new

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

$\phi_0 = c$, $u_t = \varepsilon_t$

whaaaaat

AR(1)

$$Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t$$

$\phi_1 \to 0.9$

depends

Assume

$u_t \sim iid(0, \sigma_u^2)$

$\sigma_u^2 \to$ a number

① $\mu = E(Y_t) = \phi_0$

② $V(Y_t) = \gamma_0 = E(Y_t - \mu)^2$

$= (1 + \theta_1^2)\sigma_u^2$

$-1 \leq \rho_k \leq 1$

② $k=0 \Leftarrow \gamma_k = E\big((Y_t - \mu)(Y_{t-k} - \mu)\big)$

γ_0

2a. $k=1$ $\gamma_1 = \theta_1 \sigma_u^2$

$k \geq 2$ $\gamma_2 = 0$

$\gamma_3 = 0$

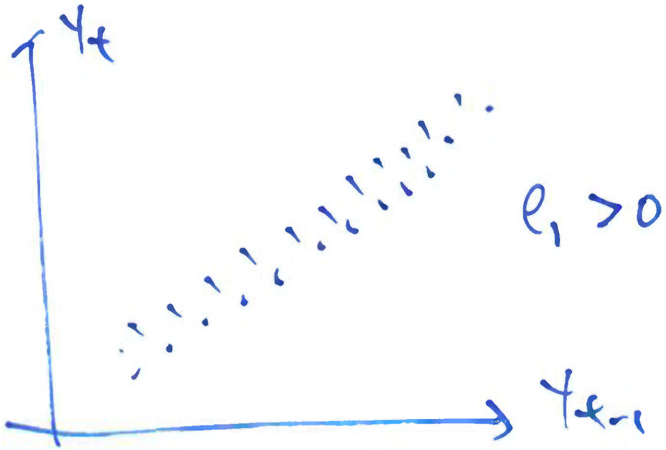
$\gamma_4 = 0$

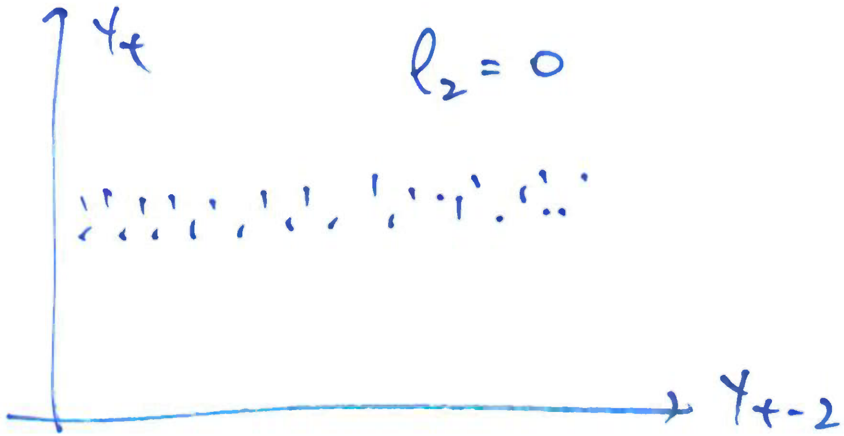
$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

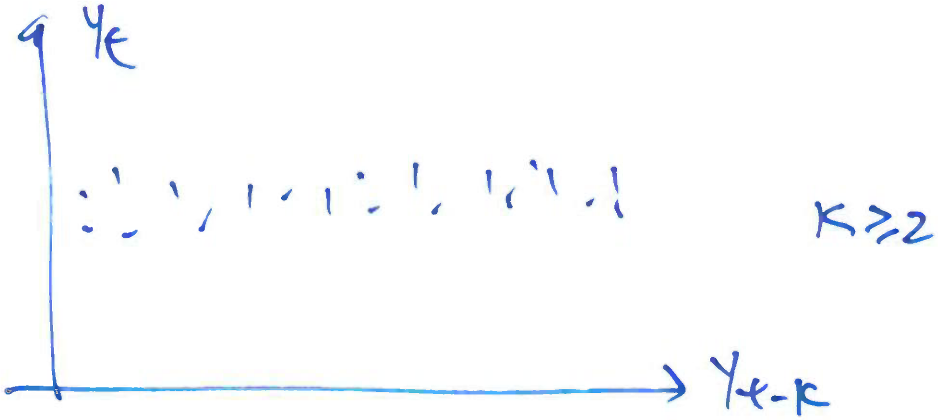
$$\rho_1 = \frac{\theta_1}{1+\theta_1^2}$$

$\rho_2 = 0$

$\rho_3 = 0$

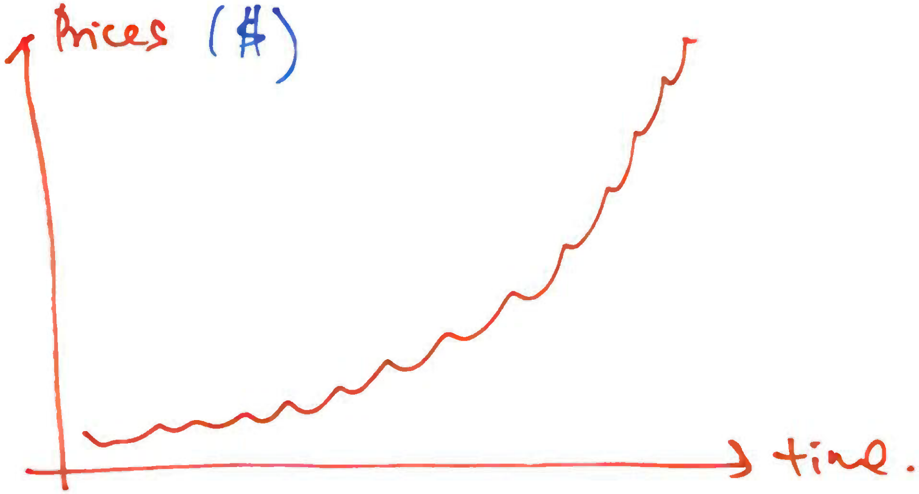




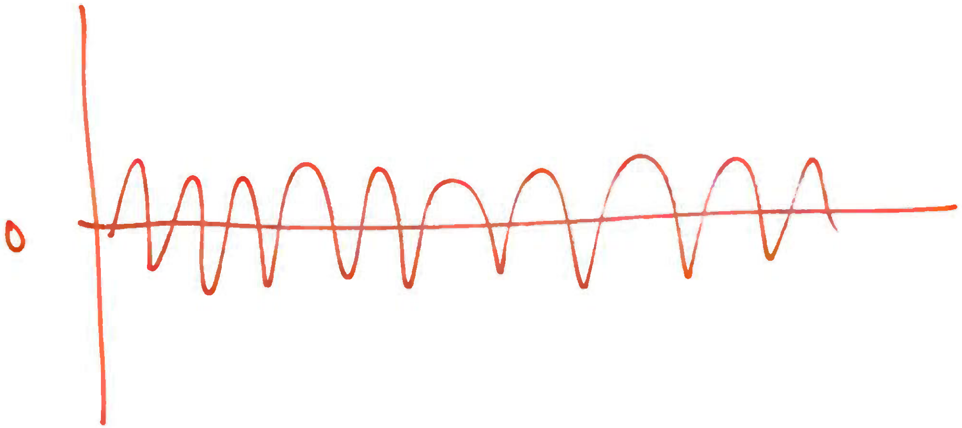


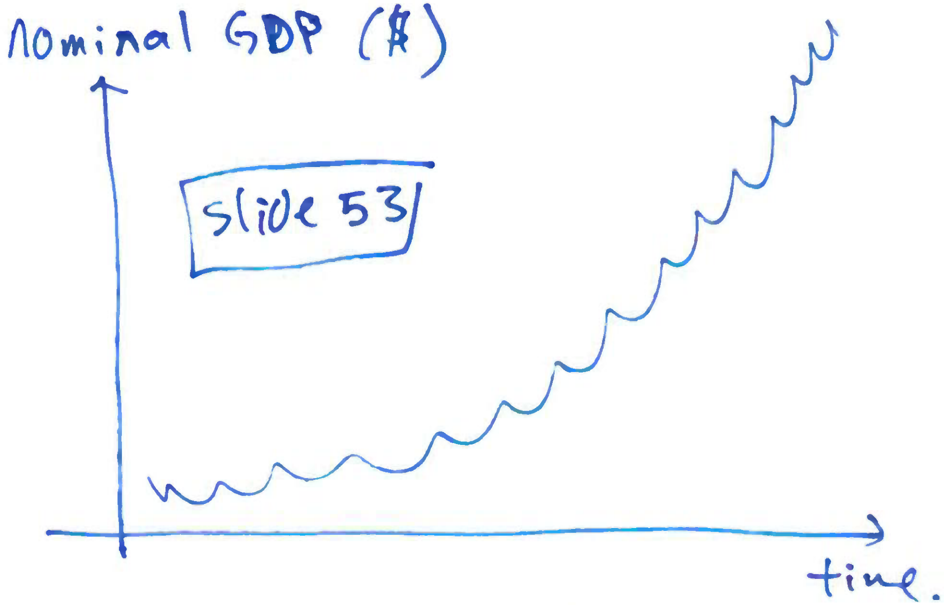
MA(1)

Week 9



$$\text{log-return} = \log \text{Prices}_t - \log(\text{Prices})_{t-1}$$





$$\underbrace{\text{real GDP}}_{\text{RGDP}} = \frac{\text{nominal}}{\text{Prices}} \Rightarrow \boxed{\text{slide 55}}$$

Prices → CPI

$$\text{growth rate} = \log(\text{RGDP}_t) - \log(\text{RGDP}_{t-1})$$